Course Code: CSE 2024	**Course Name**: Operating System	
Credits: 3 (2-0-2)	**Contact Hours:** 2-hour Theory & 2 hours lab per week	
Batch: 2024, 3rd Sem **Academic Year:** 2025-26	**Semester Duration:** 19th Jan 2026 to 8th May 2026	
Course Faculty: **1. Dr. Shilpa Mahajan** *Shilpa.mahajan@bmu.edu.in, 9882450155 and Cabin # 7, IV floor, E2 Building* **2. Dr. Sukhandeep Kaur** Sukhandeep.kaur@bmu.edu.in, 9501880466 and Cabin #12, *E2 Building* ***3. Dr. Isha Ganguli*** *Isha.ganguli@bmu.edu.in, 8481835187 and Cabin #87, E2 Building* ***4. Dr. Harsh Taneja*** *Harsh.taneja@bmu.edu.in, 7838797610 and Cabin # 7, IV floor, E2 Building*	**Course Coordinator:** Dr Shilpa Mahajan **Email:** *Shilpa.mahajan@bmu.edu.in, 9882450155 and Cabin # 7, IV floor, E2 Building*	

Aim of the course: The focus of this course is to teach the students to understand the concepts and features of the Operating System.

Course Overview and Context: The course is meant to introduce students the fundamental concepts involved in operating system and how operating systems acts as an interface between the underlying hardware and the applications programs used by the user of a computing system. An understanding of what process is, how synchronization is done and how inter-process communication takes place. This course also covers memory, file and device management. Also, to deal with protection and security issues related to operating systems.

Course Outcomes (CO): At the end of the course the students should be able to do the following:

	Course Outcome Statement	**Mapped Component**	**Associated Activities / Assessment**
CO 1	Understand how application software interacts with hardware with the help of operating system	Employability	→ Intensive theory sessions. → In-class tool Information → Hands-on practical sessions in labs.
CO 2	Implement various CPU scheduling algorithms	Skill Development	→ Hands-on Algorithm Simulation → Group problem-solving sessions → Programming assignments
CO 3	Analyze operating system with respect to process management, memory management etc.	Skill Development	→ Analytical assignments → Tutorial sessions on process states
CO 4	Evaluate the performance and features of current operating systems being used in industry.	Entrepreneurship, Employability	→ Theory Sessions → Industry Tool Exposure

Prerequisite: if any

1. None

Topics of the course:

Topics	Number of Sessions (48)	Course Outcomes
Introduction to Operating system and types of operating system. Role of OS in real life applications	3(2 Th + 1 Lab)	CO1
Understand the concept of process, attributes related to process and operations performed on the process	3(2 Th + 1 Lab)	CO1
Discuss the process state diagram, role of context switching and dispatcher.	3(2 Th + 1 Lab)	CO1, CO2
Discuss why we need to perform CPU scheduling and under which condition this scheduling is done and goals related to it.	2(1 Th + 1 Lab)	CO2
Explain the different CPU Scheduling Algorithms: First Come First Serve, Shortest Job First, Round Robin Scheduling, Longest Job First, Highest Response Ratio Next, Priority Based Scheduling, Multilevel Queue Scheduling, and Multilevel Feedback Queue Scheduling	4(2 Th + 2 Lab)	CO2
Understand how process synchronization is done and discuss what kind of problem arises not having synchronization between processes, conditions to be followed to achieve synchronization and solutions.	4(2 Th + 2 Lab)	CO2
Familiarize with the concept of concurrent programming	3 (2 Th +1 Lab)	CO2
Discuss characteristics, prevention, avoidance, detection and recovery of deadlock	3 (2 Th +1 Lab)	CO3
Explain thread, its advantages and categorization	2 Th	CO3
Discuss the contiguous and non-contiguous memory management techniques	3 Th	CO3, CO4
Familiarize with the concept of Virtual Memory and how it can be implemented	2 Th	CO3
Discuss the page replacement algorithms: First In First Out, Optimal Page Replacement, Least Recently Used and Most recently used	3(2 Th + 1 Lab)	CO4
Discuss the role of file management, directory structure, operation and disk space allocation method.	2 Th	CO4
Explain the various disk scheduling algorithms.	4(2 Th + 2 Lab)	CO4

CO/PO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3	PSO 4
CO1	1	3			3								2		1	

CO2	1	2	1		1								1		2	
CO3		1			2								1		2	
CO4		1	2	2	2					1	2		1		2	

Learning Resources:

Textbook:

1. Operating Systems Concepts – A. Silberschatz, P. Galvin and G. Gagne. Wiley India [Complete syllabus (100%) is being covered from the textbook]

Refernce Books:

1. Operating Systems Concepts - Gary Nutt, N. Chaki and S. Neogy, Pearson Education
2. Operating Systems – W. Stallings, Pearson Education
3. Operating Systems: A Concept-based Approach – D. M. Dhamdhere, Tata McGraw-Hill

Video lectures/ MOOC:

Reference Links:
Due to the ever-changing nature of the web, current material that is freely available will be relied upon, including software documentation, blog posts, tutorials, and more. Links to the readings will be provided on a weekly basis in addition to the following online course:
NPTEL Operating Course Link:
https://www.youtube.com/watch?v=jciGIvn7UfM&list=PLvqSpQzTE6M9SYI5RqwFYtFYab94gJpWk

Experiential Learning Component:
The Experiential Learning Component of the Operating System course (CSE2024) is designed to provide students with hands-on experiences that reinforce theoretical knowledge through practical application. Throughout the course, students will engage in project-based learning where they will be assigned practical projects that evolve from basic Shell scripting and other scheduling algorithms. These projects, which span the entirety of the course, are structured into phase-wise evaluations to ensure consistent application and progression of skills. The first project phase evaluation, scheduled for the week starting 16th September with a weightage of 30%, focuses on the fundamentals of Linux commands and shell scripting, assessing students on designing algorithms, learning about scheduling, accessing process, and memory management. The second project phase, starting the week of 4th November and also weighted at 30%, emphasizes implementation of Ionic framework, page fault algorithms, CPU scheduling (App), page fault (app), writing a mini shell, and basic boot loader.

Assessment Pattern: The final grade will be determined by the marks or grades earned during the project's phase-wise evaluations and the end-term assessment. Grading will be conducted using the relative grading method outlined in the university's academic regulations. To be eligible for grading, students must achieve a minimum of 40% of the total marks upon completing all assessments listed in the table below:

Evaluation Component	Weightage (%)	Evaluation Schedule	Rubrics	CO Mapping
Quiz (2)	30%	March (3rd Week) and May (3rd Week)	– Topics to be covered will be announced in the class. – Topics to be covered will be announced in the class. – Mode of quiz will be offline.	CO1, CO2 CO3

Lab Evaluation	30%	Continuous	Implementation of practical learning and Lab Viva (20%) Problem solving and implementation (10%)	CO1, CO2, CO3
End Term Evaluation	40 %	As per academic calendar/Date-sheet	Close book written exam	CO1, CO2, CO3 and CO4

- All evaluations will be based on the work presented by the students as well as the questions asked, or the problems given to code.
- Cases of AI-generated code or plagiarism will be taken seriously and reported according to the university's policy on Unfair Means (UFM). It is essential that all work is original and adheres to academic integrity.
- There is a mandatory requirement to upload the project to a public repository on GitHub.

Student Responsibilities:

- Attend lectures and do the work Lab Assignments as per instructions.
- Participate in the discussions/assignments held during classes.
- Check announcements on the LMS and emails regularly.
- Submit the assigned task on time.
- Regularly check marks on the LMS to ensure they are up to date.
- Participate in class and take necessary actions to grasp the material. Asking questions is encouraged.
- Communicate any concerns by speaking directly with the instructor.

Attendance Policy: Students are expected to attend classes regularly. Failure to follow the classes regularly and adhere to the expected attendance percentage will result in losing quiz/lab marks and a reduction of the grade as per the University's grading policy.

Recourse Examination Policy: In case a student fails the course, a one-time recourse is permitted as per the academic regulations of the University. Recourse is allowed only for the End Semester examination.

Make-up policy: No make-up exam will be conducted for unexcused absences. The faculty needs to be informed in advance in case the student is not going to appear for any evaluation component, and it is at the discretion of the faculty to sanction makeup for an evaluation component.

Behavior Expectations: No mobile phones and other distractive gadgets are permitted in the class.

Academic Dishonesty/Cheating/Plagiarism: Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.